Coupled Pendulum

Name: Vedanta Mohapatra

Section-1, Group -1

Roll No: 20CS02001

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Aim of the Experiment

To determine the resonance and beat time period of a coupled pendulum.

Apparatus Required

(i) Coupled pendulum set up (connected to computer and printer).
(ii) Stop watch.

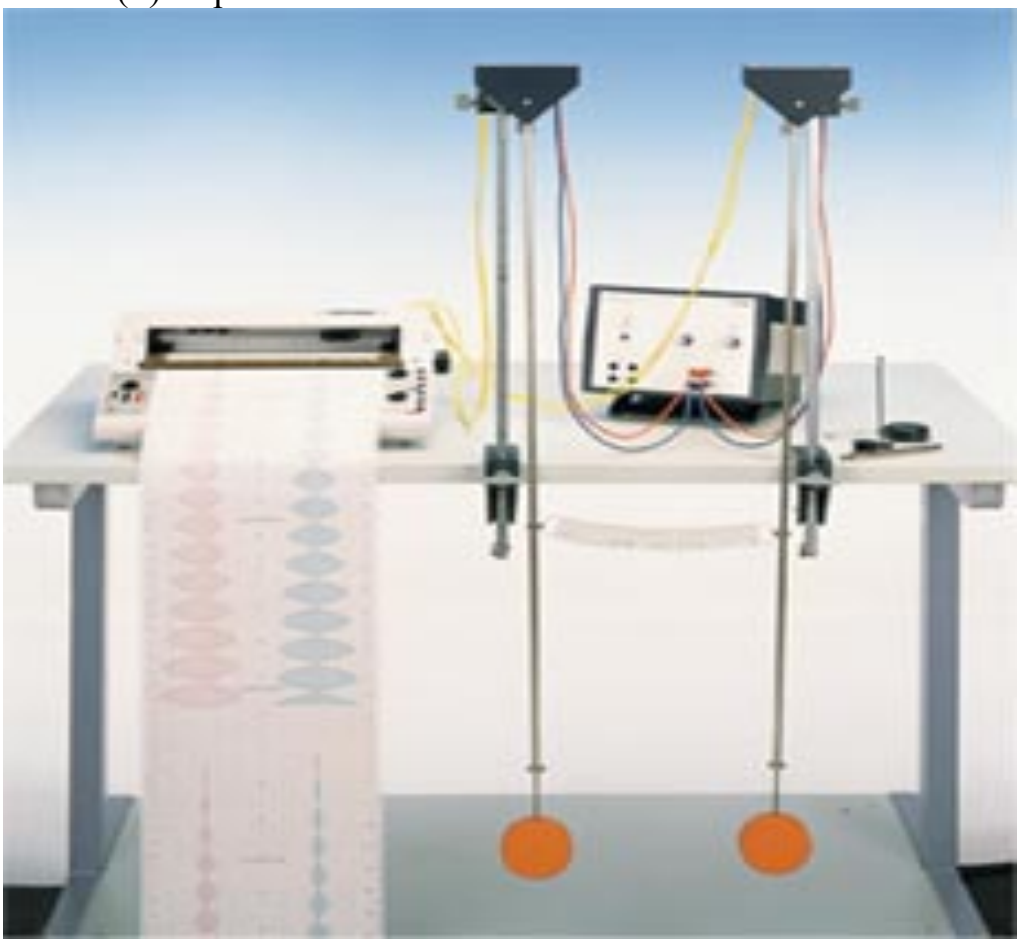


Fig-1: Experimental set up

Theory

The experimental set up is shown in Fig-1. It comprises of a coupled pendulum having two identical pendula each with a rod of length 'l' supporting a mass 'm' coupled by a presumably massless spring of stiffness 'k'. A schematic representation of the set up is shown in fig-2.

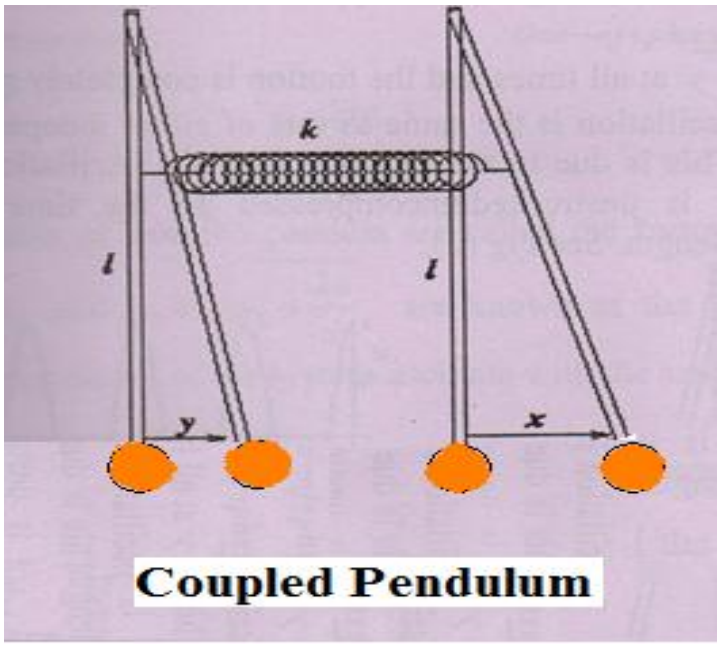


Fig-2: Schematic diagram

Here, the length of the spring is the separation between the two pendulums at zero displacement as shown in Fig-2.

If x and y are the respective displacements of the two masses, then the equation of motion for the coupled pendulum becomes

$$m\ddot{x}=-mg\frac{x}{l}-k(x-y) \quad \text{and} \quad m\ddot{y}=-mg\frac{y}{l}-k(y-x) \qquad (1)$$

Writing the natural frequency of each pendulum as $\omega_0^2=\frac{g}{l}$, *we have*

$$\ddot{x}+\omega_0^2x=-\frac{k}{m}(x-y) \quad \text{and} \quad \ddot{y}+\omega_0^2y=-\frac{k}{m}(y-x) \qquad (2)$$

Adding Eqns. (1) and (2) we get,

$$\ddot{x}+\ddot{y}+\omega_0^2(x+y)=0 \qquad (3)$$

And subtracting Eqn. (2) from Eqn. (1) we get,

$$\ddot{x} - \ddot{y} + \omega_0^2(x-y) + \frac{2k}{m}(x-y) = 0 \quad \text{.....} \quad (4)$$

Substituting X= x+y and Y=x-y in (3) & (4) respectively, we obtain the following equations

$$\ddot{X} + \omega_0^2 X = 0 \quad \text{.....} \quad (5).$$

$$\ddot{Y} + \left(\omega_0^2 + \frac{2k}{m}\right) Y = 0 \quad \text{.....} \quad (6)$$

The Eqns. (5) and (6) are the equations of simple harmonic motion with natural angular frequencies ω_0 and $\omega_1 = \sqrt{\left(\omega_0^2 + \frac{2k}{m}\right)}$ respectively.

<u>Case-I (In-Phase mode)</u>

When x=y and Y=0, then Eqn. 5 is valid. This case is equivalent to the uncoupled case. The spring is unstressed and uncompressed when the coupled pendulum is in phase. In this case, the two pendulums oscillate in the same direction with equal amplitudes.

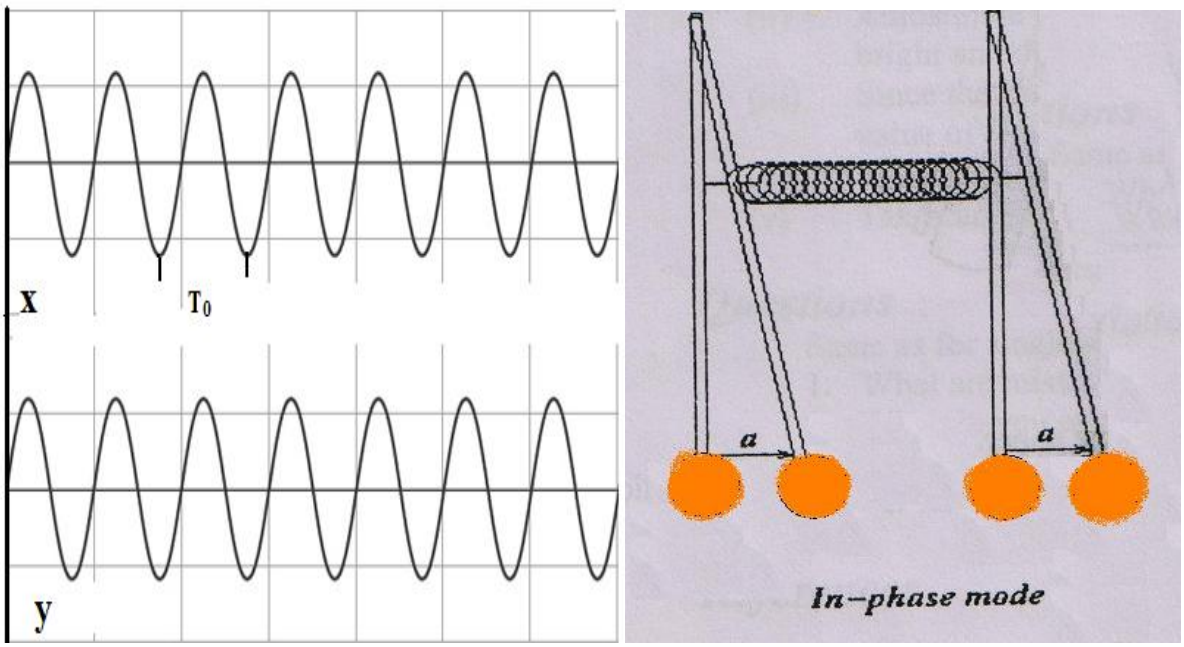


Fig-3: In phase mode

<u>Case-II (Out of Phase mode)</u>

When x= - y and X=0 Eqn. 6 is valid to represent the oscillation of the pendulum. The frequency of oscillation is greater than the natural frequency of the

independent pendulum as the spring is either compressed or stretched. In this case, one pendulum oscillates in just the opposite direction to that of the other and this is called as out of phase mode of oscillation.

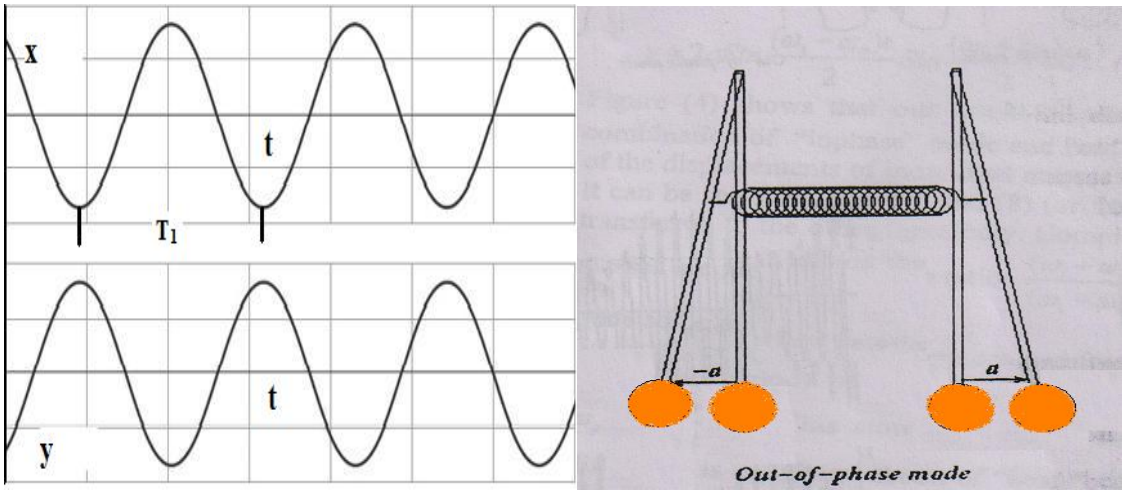


Fig-4: Out of Phase mode

Case-III (Resonance)

When the natural frequency of any system matches with the frequency of the force applied, *resonance* occurs. In case of resonance, x=2a or y= -2a.

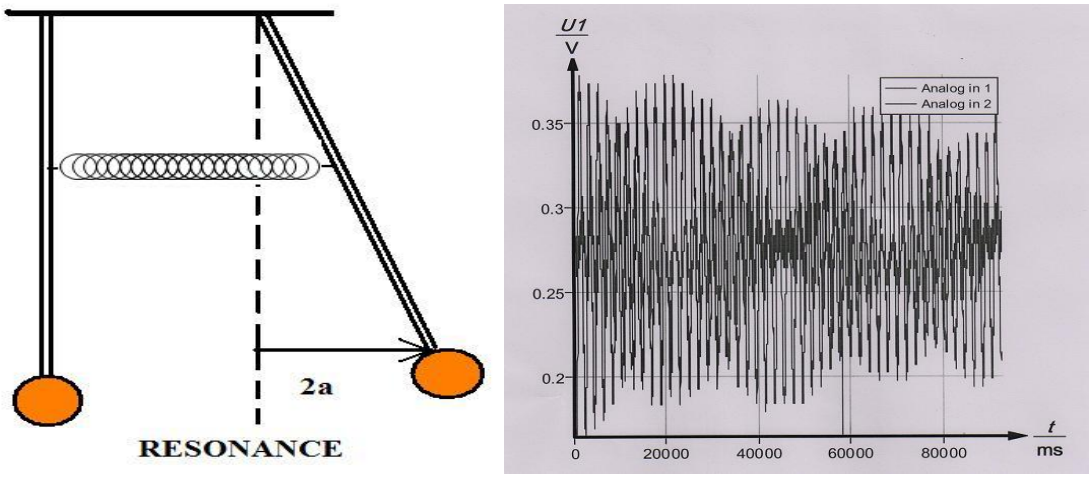


Fig-5: Resonance/Beat mode

So, the general solution for Eqn. 5 and 6 are the following:

$$X = x+y = X_0 \cos(\omega_0 t+\phi_1)$$

$$Y = x-y = Y_0 \cos(\omega_1 t+\phi_2)$$

Where $\omega_1^2 = \omega_0^2 + \frac{2k}{m}$. *For resonance we choose amplitude* $X_0 = Y_0 = 2a$ *and phase* $\phi_1 = \phi_2 = 0$, *the displacement of right pendulum is given by:*

$$x = \frac{1}{2}(X+Y) = a \cos \omega_0 t + a \cos \omega_1 t$$

$$= 2a \cos\frac{(\omega_1-\omega_0)t}{2} \cos\frac{(\omega_1+\omega_0)t}{2}$$

Similarly the displacement of left pendulum is

$$y = X-Y = a \cos (\omega_0 t) - a \cos (\omega_1 t)$$

$$= 2a \sin\frac{(\omega_1-\omega_0)t}{2} \sin\frac{(\omega_1+\omega_0)t}{2}$$

The displacement given to one of the pendulum as, x=2a, y=0 at t=0 executes a motion which is a combination of 'in-phase' and 'out of phase' in equal proportion. Energy of one pendulum (oscillator) gets transformed to the other as they oscillate. When $W_1=W_2$; (where W_1 and W_2 are the weights of the two pendulums) and $\frac{\omega_1+\omega_0}{\omega_1-\omega_0}$ is an integer, complete energy exchange occurs and the two pendulums exhibit successive stops and maximum displacement. The pendulum which stops has zero energy and the other one is in maximum energy state.

Since in phase and out of phase modes are combined, the plot we get is a result of two frequencies, one with a high frequency $\omega_c = \frac{\omega_1+\omega_0}{2}$(resonance frequency) and other with a lower frequency. The second one with a lower frequency occurs at half of the normal mode frequency difference whose value is $\frac{\omega_1-\omega_0}{2}$ and the phenomenon called as beat. This occurs when $\omega_1 \simeq \omega_2$. The amplitude of oscillation varies with this frequency and we get the envelope pattern as shown in the Fig-6.

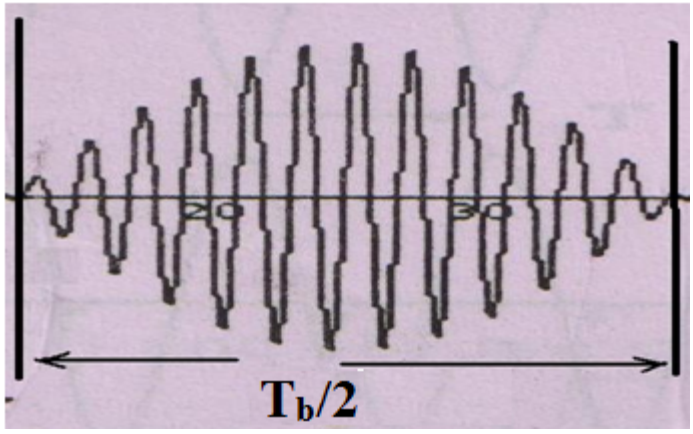


Fig-6: Envelope pattern amplitude vs frequency.

Thus, we have $T_0 = \frac{2\pi}{\omega_0}$ (In phase time period)

$T_1 = \frac{2\pi}{\omega_1}$ (Out of phase time period)

$T_c = \frac{4\pi}{\omega_1+\omega_0} = 2\frac{T_0T_1}{T_0+T_1}$ (Time period for coupled case or resonance time period)

$T_b = \frac{4\pi}{\omega_1-\omega_0} = 2\frac{T_0T_1}{T_0-T_1}$ (Time period for beat)

The degree of coupling is $\chi = \frac{\omega_1^2-\omega_0^2}{\omega_1^2+\omega_0^2} = \frac{T_0^2-T_1^2}{T_0^2+T_1^2}$

Procedure

(i) **Uncouple** the pendulums.

(ii) Take three independent observations of each pendulum (***50 oscillations each***).

(iii) Calculate the **time period** for one oscillation taking the average over the three oscillations. This is the time period in the uncoupled case.

(iv) Now **couple** the two pendulums with the given spring.

(v) Impel both the pendulums to the same extent in same direction and calculate the ***time period*** of the oscillation (*called* ***in-phase time period*** $\boldsymbol{T_0}$) by taking *three* independent readings each for 50 oscillations.

(vi) Impel the two pendulums to the same extent but in **opposite** direction. Calculate the **time period** of oscillation by taking three independent readings each for 50 oscillations (*called the* ***out of phase time period*** $\boldsymbol{T_1}$).

(vii) Impel one of the pendulums to some extent keeping the other one fixed. This is called as "**resonance mode**". Both the pendulums resonate with total energy E where $E=E_1+E_2$. The value of E_1 and E_2 varies from zero to E.

(viii) When $E_i = 0$ (i = 1 or 2), it constitutes a stop of the pendulum. Observe the time period for five consecutive stops of a particular pendulum. Calculate the time period averaged over three independent observations. This is called ***beat time period***.

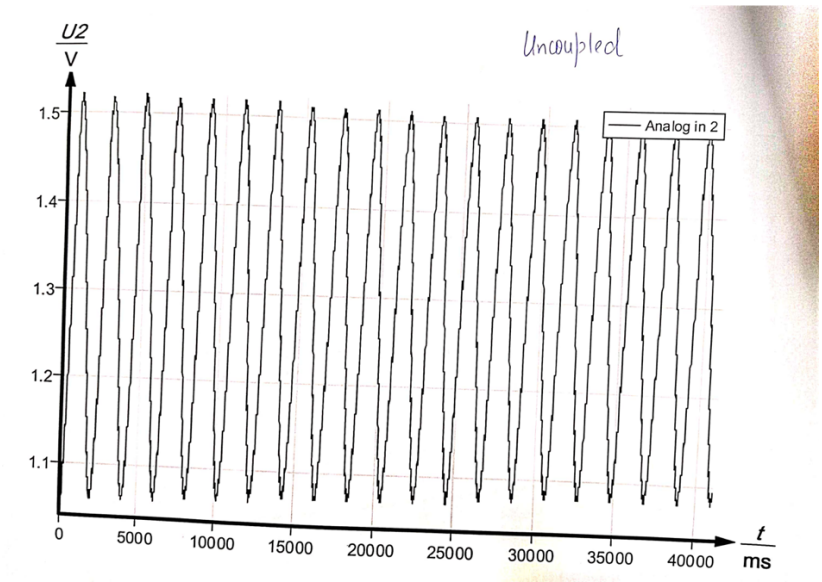
(ix) When the coupled pendulum oscillates in the resonance mode, observe individual pendulums. Take three readings of time for five oscillations of each pendulum. Calculate the time period by averaging over the total numbers & number of oscillations. This is called the ***resonance time period.***

(x) Now compare the values $T_{measured}$, $T_{computational}$ and $T_{calculated}$ and calculate the percentages of error Where $T_{measured}$ is the T value measured directly. $T_{calculated}$ is the 'T' value calculated from the equation. $T_{computational}$ is the 'T' value measured from the computer printed graph.

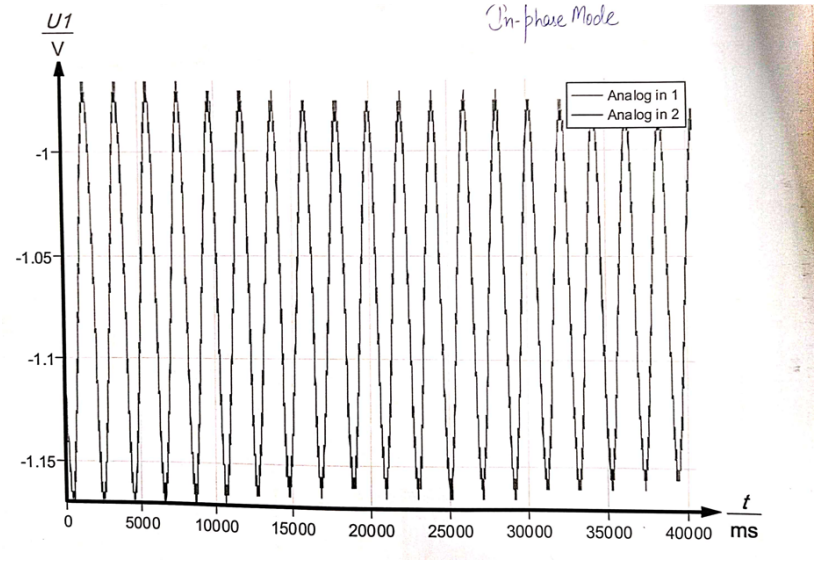
Observations

Table-1: Resonance and beat time period.

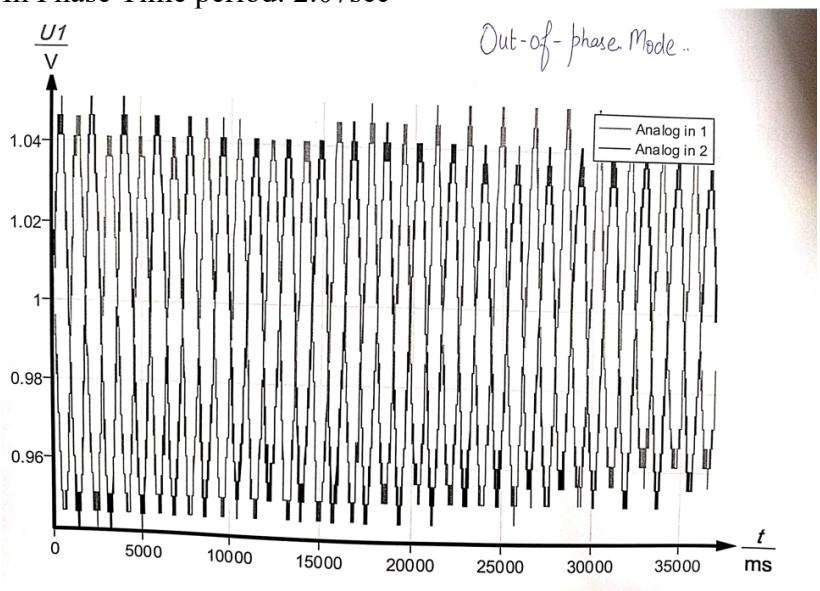
UNCOUPLED		COUPLED							
Time for 20 oscillations	Average T	IN-PHASE		OUT OF PHASE		BEAT		RESONANCE	
		Time for 20 Oscillations		Time for 20 Oscillations		Time for 5 stops in secs		Time for 5 Oscillations in secs	
t_1= 40.39	Avg. T= $\frac{t_1+t_2+t_3}{60}$ =2.04	t_1= 41.25	Avg. T_0= $\frac{t_1+t_2+t_3}{60}$ =2.06	t_1= 37.64	Avg. T_1= $\frac{t_1+t_2+t_3}{3}$ =1.88	t_1= 116	Avg. T_b= CT_b=Beat Time period= $\frac{t_1+t_2+t_3}{3*2.5}$ =46.67	t_1= 9.53	T_c= (Resonance Time period) $\frac{t_1+t_2+t_3}{15}$ =1.88
t_2= 41.11		t_2= 40.89		t_2= 37.78		t_2= 118		t_2= 9.41	
t_3= 40.99		t_3= 41.34		t_3= 37.69		t_3= 116		t_3= 9.28	



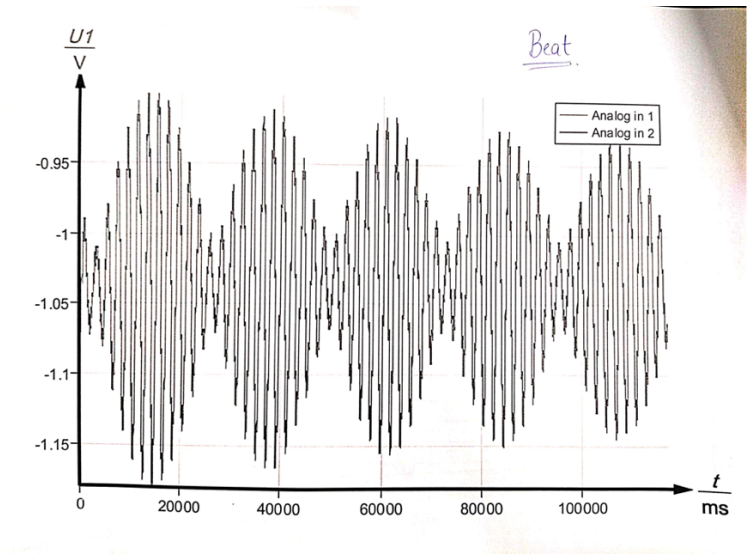
Uncoupled pendulum T = $\frac{40000.10^{-3}}{19.25} = 2.08$



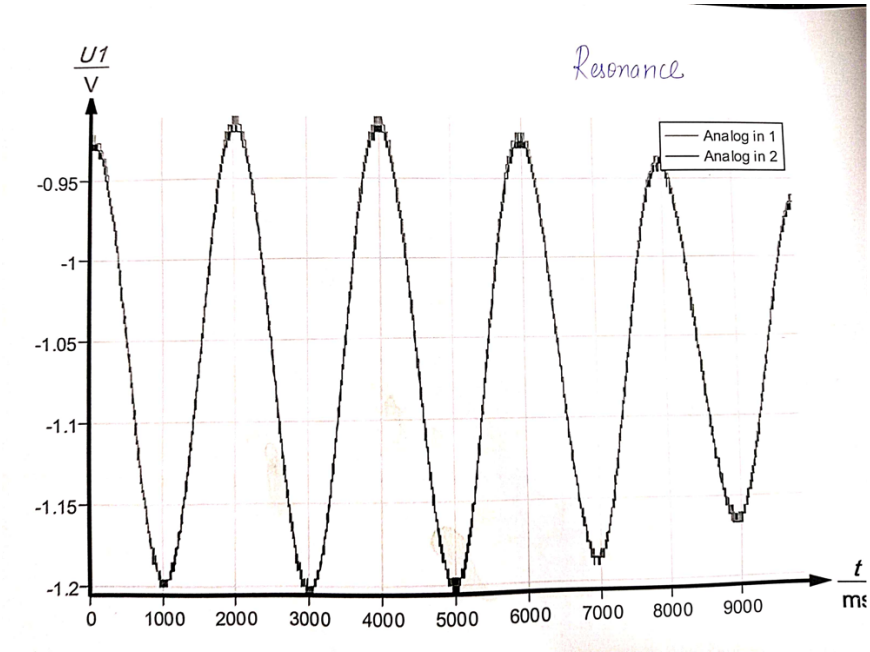
In Phase Time period: 2.07sec



Out of Phase Time Period = 1.818sec



Beats Time Period = 46.5 sec



Resonance Time Period = 2 sec

Calculations:

$k = 1\ Nm^{-1}$; $l = 1m$; $g = 9.8\ m/s^2$; $m = 1kg$

We know that $\omega_o = \sqrt{\frac{g}{l}} = \sqrt{9.8} = 3.13$ rad/sec (ω_o is natural frequency of pendulum)

For coupled oscillations: $\omega_1 = \sqrt{\omega_0^2 + \frac{2k}{m}} = \sqrt{9.8 + 2} = 3.43$ rad/sec

In Phase Time Period $T_o = \frac{2\pi}{\omega_o} = \frac{6.28}{3.13} = 2.006$ sec

Out of Phase Time Period $T_1 = \frac{2\pi}{\omega_1} = \frac{6.28}{3.43} = 1.8309$ sec

Beat Time Period $T_b = \frac{4\pi}{\omega_1 - \omega_o} = 41.86$ sec

Resonance Time Period $T_c = \frac{4\pi}{\omega_1 + \omega_o} = 1.91$ sec

Results:

Type of Oscillation	$T_{measured}$	% Error ($T_{measured}$)	$T_{computational}$	% Error ($T_{computational}$)	$T_{calculated}$
Uncoupled Oscillation	2.04	1.69%	2.08	3.69%	2.006
In-Phase Oscillation	2.06	2.69%	2.07	3.19%	2.006
Out of phase Oscillation	1.88	2.68%	1.818	-0.70%	1.8309
Beat Oscillation	46.67	11.49%	46.5	11.08%	41.86
Resonance Oscillation	1.88	-1.57%	2	4.71%	1.91

where,

% Error ($T_{measured}$) = $\frac{T_{measured} - T_{calculated}}{T_{calculated}} * 100$

% Error ($T_{computational}$) = $\frac{T_{computational} - T_{calculated}}{T_{calculated}} * 100$

Conclusions

(i) The time periods obtained with physical measurements agree closely with the theoretical values predicted for them with small errors.

(ii) (ii) Modelling physical phenomenon is always a business of approximations. To that end ,we have employed various approximations (small angle) and neglected certain effects(air drag and mass of spring) and the method is not exactly applicable to various real world settings but it serves as an outstanding proof of concept of simplistic approaches to analysing complicated systems in physics and elsewhere

Precautions

(i) The independent pendulums should be adjusted first with the help of moving masses to have approximately equal oscillation time periods.

(ii) Amplitude of oscillations should be kept small i.e. the angular amplitude should be kept below 10^0.

(iii) Time should be noted for large number of oscillations.

(iv) Spring should be kept at its natural length. It should not be too loose otherwise sag will be produced in the string.

Considerations

(i) What is simple harmonic motion and a simple pendulum? Find out the time period of a simple pendulum.

A simple pendulum is a bob that is suspended from a rigid support which oscillates due to the restoring force of gravity when displaced from its mean position

(ii) What is coupled pendulum and define its spring constant.

Two (or more) pendula connected to each other by using springs. The stiffness of the spring is defined as the spring constant of the coupled pendulum. It characterizes how strongly(in terms of force) one pendulum affects the other's motion.

(iii) What are normal modes of vibration and resonance?

The normal modes of vibration are

• In phase mode: The instantaneous displacements of both the bobs are equal to each other.

Mathematically, $x(t)= y(t)= \varphi\cos(\omega_0 t+\theta)$. Both pendulums oscillate in phase with their natural frequencies. This is due to the fact that there is no relative displacement between the two bobs.

• Out of phase mode: The instantaneous displacements of both the bobs are equal and opposite to each other. Mathematically, $-x(t)=y(t)=\varphi\cos(\omega_0 t+\theta)$.Both pendulums oscillate out of phase with each other with the frequency $\omega_1 = \sqrt{\omega_0^2 + \frac{2k}{m}}$

• Resonance: When one bob is held constant and the other is displaced about the mean position .Then the moving bob exerts a periodic force with the same frequency as the natural frequency of

oscillation on the other bob ,resonance is achieved and maximum transfer of energy takes place. Gradually the moving bob slows to a halt and the former bob oscillates with full energy. And the cycle repeats.

(iv) What is beat and beat time period?

Beat is the phenomenon of periodic variation in the amplitude of oscillation of a body(or a wave). Beat time period is twice the duration of time between two successive maxima in amplitude.

(v) How many normal modes are there and what are they?

Normal modes are the modes of oscillation where the whole system oscillates with the same frequency .There are two normal modes: in phase and out of phase. Mathematically, any oscillation of the system can be expressed as the linear combination of the normal modes.